

Report 01

MediCamp

A Community Healthcare Camp Management System

Prepared for

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1. Reason for Project Selection

1.1 Problem Statement

Free medical camps run by NGOs are a key way rural and underserved communities in Bangladesh access healthcare. Most camps still rely on paper: hand-written registration forms, paper prescriptions, and manual stock counts.

This causes real problems. Patients lose continuity of care, since doctors cannot see prior diagnoses or allergies from earlier camps. Pharmacists run out of stock without warning, or waste unused medicine. Referrals and follow-ups are rarely tracked end-to-end, and area-wise reports for donors take days to compile by hand.

1.2 Proposed Solution

We propose MediCamp, a role-based web app covering the full camp life cycle. It supports six actors — Admin, Host, Doctor, Volunteer, Pharmacist, and Patient — each with a dashboard scoped to their job.

It models the camp as a ten-phase workflow, from planning and staffing through registration, screening, consultation, dispensing, referral, follow-up, and reporting. A Blood Donation module and a multi-level (district to village) Reporting module sit on top of the same patient and camp data.

1.3 Goals

- Give each role its own login and dashboard, scoped to only the actions that role needs.
- Keep one continuous patient history across camps, instead of resetting records each time.
- Auto-deduct medicine stock on dispensing and flag shortages before they disrupt a camp.
- Track referrals and follow-up status so no patient's care is left unresolved.
- Produce area-wise and category-wise reports for Hosts, donors, and Admins.
- Link a blood donor registry and request workflow directly to patient records.

1.4 Justification of the Approach

The system's data is highly relational: patients link to visits, visits to diagnoses and prescriptions, prescriptions to inventory, and camps to staff and budgets. A change in one place must stay consistent everywhere else.

This favours a normalised relational database with foreign keys, transactions, and triggers over a loosely-structured NoSQL store, since the database itself can then enforce integrity (e.g., stock never goes negative).

This also matches the course focus: CSE 3224 emphasises database-driven information system design, so a full SQL query bank (joins, views, triggers, procedures) plus a role-based MVC layer covers both halves of the lab.

2. Background Study

Free health camps are a long-standing model used by charities, civic groups, and NGOs across Bangladesh. They are typically short, staffed by volunteer doctors, and run on a fixed donation budget — very different from a permanent hospital.

General-purpose hospital systems and open-source EMR/HMIS platforms are built for permanent facilities with continuous staffing and IT support. They do not model a temporary, budgeted, one-day event well, and rarely include a blood-donation registry.

In Bangladesh, digital-health efforts have mostly focused on tele-consultation and hospital records, not on field camps. NGOs are left with paper or spreadsheets, which cannot enforce data integrity or produce area-wise reports easily.

This confirms a gap: a purpose-built, camp-centric system that treats each camp as a budgeted, staffed event while keeping one longitudinal patient record is missing from both hospital software and ad-hoc paper processes.

3. Requirement Analysis

3.1 Functional Requirements

Functional requirements are grouped by the six actors, plus the Blood Donation and Reporting modules that sit on top of camp and patient data.

3.1.1 Admin

- Create and manage user accounts, and assign roles and permissions.
- Approve or reject Host (NGO) registration requests before camps can be created.
- Maintain the location hierarchy: districts, upazilas, unions, and villages.
- Maintain the master medicine list, hospital list, and blood-group reference list.
- Monitor all camps system-wide and generate global, cross-Host reports.
- Configure system-wide settings such as token-numbering and report parameters.

3.1.2 Host (*Organising NGO*)

- Create a camp with name, type, location, venue, schedule, expected patients, and budget.
- Assign doctors, volunteers, and pharmacists to a specific camp.
- Allocate medicines, equipment, and first-aid supplies from available inventory.
- Publish the camp once staffing and allocation are complete.
- Track camp progress in real time and manage expenses against the budget.
- View reports limited to the camps this Host has organised.

3.1.3 Doctor

- View assigned camps and the live patient queue for the active camp.
- Review a patient's visit history, past diagnoses, prescriptions, and allergies.

- Enter symptoms, diagnosis, clinical notes, and advice for the visit.
- Create a prescription (medicine, dosage, frequency, duration) sent to the pharmacy.
- Flag a follow-up requirement, or refer the patient to a hospital with urgency and notes.

3.1.4 Volunteer

- Search for an existing patient, or register a new one with personal and medical details.
- Generate a queue token and place the patient in the waiting queue.
- Record initial screening: vitals, BMI, blood pressure, and presenting symptoms.
- Conduct follow-up visits and record status (recovered, improving, missed, etc.).

3.1.5 Pharmacist

- View incoming prescriptions from the consultation phase.
- Dispense medicines, which automatically decreases camp inventory.
- Mark a medicine out-of-stock and suggest a clinically suitable alternative.
- View inventory levels and submit restocking requests to the Host.

3.1.6 Patient

- Register an account and view upcoming camp schedules in their area.
- Register for a camp and view their own visit, prescription, and referral history.
- Update their own profile information.
- Opt in as a blood donor, search donors, or submit a blood request.

3.1.7 Blood Donation Module

- Let a patient register as a donor with blood group, weight, and availability.
- Let any patient create a blood request with group, units, hospital, and urgency.
- Search donors by blood group and location, returning contact details.
- Maintain a donation history log of donor, receiver, date, and units donated.

3.1.8 Reporting Module

- Camp dashboard: patient counts, staff counts, medicines used, and wait times.
- Disease report: disease-wise counts, most common and seasonal diseases.
- Medicine report: usage, remaining stock, and out-of-stock items.
- Staff reports: patients seen, referral rate, and follow-up completion.
- Financial report: budget versus actual cost across spending categories.
- Area-based reports at district, upazila, union, and village level.

3.2 Non-Functional Requirements

Beyond the functions above, these quality attributes are needed for the system to work reliably in a real field-camp setting.

Attribute	Requirement
Security and Privacy	Role-based access must stop, e.g., a Pharmacist editing diagnoses. NID and health data need access logging.
Availability	Camps often have weak internet; the app should tolerate brief drops and retry critical writes.
Usability	Volunteer and pharmacist screens must suit low computer literacy: large buttons, minimal fields.
Data Integrity	Stock updates and prescriptions must use transactions so a failure mid-step leaves no bad state.
Performance	Token generation and stock checks should respond within one to two seconds under load.
Scalability	Schema and reports must stay fast as camps, patients, and visit history grow over time.
Maintainability	Modules and the SQL query bank should be documented so future teams can extend them easily.
Portability	The system should run on low-cost hardware and free-tier hosting for resource-limited NGOs.

4. Feasibility Analysis

4.1 Technical Feasibility

The system is technically feasible with tools the team already knows. The data layer maps cleanly onto a relational database using a normalised, 22-table schema covering location, camp, patient, clinical, and pharmacy data.

The application layer can use a mainstream MVC framework such as ASP.NET Core, which the team has working experience with, including Identity-based auth and Docker deployment.

Free-tier infrastructure is enough for academic scale: a managed Postgres instance (Supabase or Neon) plus a free Docker-based web host (Render) cover the project's deployment needs.

4.2 Operational Feasibility

The workflow mirrors how camps already run, so adoption only means recording the same steps digitally instead of on paper — no process change is required from organisers.

The main risk is digital literacy and weak connectivity in the field. This is reduced by keeping each role's screen narrow in scope, with simple forms and sensible defaults.

Training a small camp staff on one focused screen each is realistic in a short pre-camp orientation, similar to how NGOs already brief volunteers today.

4.3 Economic Feasibility

As an academic project, the main cost is development time, since all required tools (VS Code, .NET SDK, Docker, PostgreSQL) are free and open-source.

For a real NGO deployment, ongoing costs stay low: free-tier hosting can support the data volume of periodic community camps for most organisations.

Cost Item	Current Process (Paper)	Proposed System
Development / setup	None	One-time team effort
Hosting / infrastructure	Printing, paper, storage	Free-tier hosting, low recurring cost
Medicine wastage	High – manual counts	Reduced – auto-tracked stock
Donor reporting	Manual, slow	Generated on demand

These benefits — less wastage, faster reporting, and a closed referral loop — are likely to outweigh the small infrastructure cost, making the project economically justifiable.

5. Sustainability (Aligned SDGs)

The system is directly aligned with several United Nations Sustainable Development Goals (SDGs), summarised below.

SDG	Alignment
SDG 3 – Good Health and Well-being	Makes free camps more reliable, from diagnosis through follow-up, improving access to essential care.
SDG 10 – Reduced Inequalities	Area-wise reporting and continuous records target rural and low-income communities lacking regular care.
SDG 5 – Gender Equality	Pregnancy-status fields at registration and screening support better recognition of women's health needs.
SDG 17 – Partnerships for the Goals	Coordinates NGOs, doctors, hospitals, and the community, matching SDG 17's partnership model.
SDG 9 – Industry, Innovation and Infrastructure	Replaces paper logistics with low-cost digital infrastructure other NGOs can also adopt.